

GRI-Aligned Sustainability Reporting Assessment via Knowledge Graph-RAG and LLM-Based Question-Answering: A Case Study of Vietnamese Banks

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Abstract. Environmental, Social, and Governance (ESG) disclosures have become central to sustainability evaluation, risk management, and investment decision-making in the banking sector. However, ESG and financial reports are typically lengthy, heterogeneous, and semi-structured, making systematic assessment challenging. This study introduces a GRI-aligned ESG assessment system for Vietnamese banks utilizing a Knowledge Graph-enhanced Retrieval-Augmented Generation (KG-RAG) architecture integrated with Large Language Models (LLMs) for question answering (Q&A). Our system comprises three elements: (i) an LLM-guided information extraction pipeline that converts sustainability and annual reports into structured ESG statements consistent with the Global Reporting Initiative (GRI) standard; (ii) an ESG knowledge graph that explicitly represents entities, relationships, ESG categories, and GRI indicators with complete provenance; and (iii) a hybrid KG-driven retrieval mechanism that combines graph filtering, semantic embeddings, and keyword matching to facilitate multi-hop reasoning and evidence-based answer generation. The system is evaluated on a dataset of 120 ESG questions designed and annotated by domain experts, covering Environmental, Social, and Governance dimensions across Vietnamese banking reports. Experimental findings indicate that the improved KG-RAG configuration attains the highest overall accuracy of 88.14%, surpassing conventional retrieval baselines and regular KG retrieval methods. These findings demonstrate that structured graph constraints combined with LLM-based Q&A significantly improve accuracy, traceability, and consistency in ESG assessment.

Keywords: ESG · GRI Standards · Knowledge Graph-RAG · Vietnamese Banks · Large Language Model

1 Introduction

Sustainability has become a strategic concern for firms, regulators, and investors because it influences risk management, capital allocation, and long-term value

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O. Gervasi et al. (Eds.): ICCSA 2026 Workshops, LNCS 16758, pp. 100–116, 2027.
https://doi.org/10.1007/978-3-032-30497-1_7

creation. Firms that integrate sustainability into their operations are often viewed as more resilient and better positioned for sustained performance [1]. At the same time, stakeholders' demand for reliable sustainability information has accelerated the adoption of Environmental, Social, and Governance (ESG) disclosures, making ESG performance increasingly relevant to investment analysis and risk assessment [2,3]. The UN Global Compact report *Who Cares Wins* helped formalize this shift by linking ESG considerations to mainstream financial markets [4].

Global reporting frameworks have emerged to enhance comparability and transparency in sustainability disclosure. The Global Reporting Initiative (GRI)[1] provides a widely adopted standard built on the principle of *impact materiality*, requiring entities to disclose issues that significantly affect the economy, environment, and people [5]. This stakeholder-oriented design makes GRI a natural reference for banking, where ESG factors directly influence credit, market, operational, and reputational risk. International and national guidance reinforces this need: the Basel Committee on Banking Supervision [6] has issued principles for climate-related financial risks, and in Vietnam, Directive No. 03/CT-NHNN promotes green credit and environmental–social risk management [7]. Together, these developments motivate systematic, evidence-grounded assessment of sustainability disclosures. While large language models (LLMs) and retrieval-augmented generation (RAG) enable natural-language access to such reports, embedding-only approaches often struggle with the multi-hop reasoning, cross-section aggregation, and strict evidence consistency required in ESG–finance settings [11,12].

The length and heterogeneity of sustainability reports, which combine narrative text, tables, and semi-structured disclosures, make manual analysis difficult. Natural language processing (NLP) and question answering (QA) systems are therefore valuable for supporting evidence retrieval and structured comparison across firms [8,9].

To address these limitations, we present a Knowledge Graph-enhanced Retrieval-Augmented Generation (KG-RAG) architecture for document-grounded question answering over sustainability reports and financial statements of Vietnamese banks. The core idea is to build a domain knowledge graph that explicitly represents banks, reporting periods, ESG themes, targets, emissions, green credit programs, and financial indicators, and then use graph-guided retrieval to improve semantic search. This design aims to improve evidence selection for multi-hop questions, reduce hallucination through structural grounding, and increase explainability through entity-linked evidence paths. Our work is also motivated by the growing availability of ESG-oriented NLP resources and benchmarks such as FinSim4-ESG [10].

This paper makes three main contributions:

– We develop a GRI-aligned ESG information extraction and normalization pipeline that transforms sustainability and annual reports of Vietnamese banks into provenance-preserving, statement-level evidence.

[1] https://www.globalreporting.org/.

- We design and implement an ESG Knowledge Graph integrated with a hybrid KG-RAG framework that combines graph filtering, semantic vector retrieval, and keyword matching to improve multi-hop reasoning and reduce hallucination.
- We construct an expert-annotated ESG question-answering benchmark and empirically demonstrate that the enhanced KG-RAG configuration achieves high overall accuracy, outperforming strong retrieval baselines.

2 Related Work

Information retrieval (IR) is central to question answering (QA) over large document collections. Sparse lexical methods such as BM25 remain strong baselines because they are efficient and robust for keyword-oriented matching [13]. Dense retrieval methods such as Dense Passage Retrieval (DPR) address vocabulary mismatch by mapping queries and passages into a shared embedding space [14].

LLMs have recently been explored across the IR pipeline, including query rewriting, retrieval, reranking, and answer generation, offering stronger semantic modeling but also raising concerns about reliability in high-stakes settings [15,21]. Knowledge graphs (KGs) complement text retrieval by explicitly representing entities, relations, and schema constraints, thereby supporting multi-hop reasoning and structured evidence selection [17]. Recent studies further show that LLMs can help construct KGs from unstructured text through prompting-based extraction pipelines, although normalization and quality control remain important [18,20].

In the ESG domain, LLM-based approaches have been applied to both ESG evaluation and the extraction of structured signals from sustainability reports [16,23]. However, most rely on embedding-only or standard RAG pipelines and provide limited support for transparent evidence chaining across scattered disclosures. Such pipelines typically struggle with multi-hop reasoning over ESG–finance relationships, lack robust mechanisms for cross-document aggregation, and offer weak control over answer grounding, which raises hallucination risk in regulatory or investment settings. Auditable ESG assessment therefore calls for a graph-aware retrieval strategy that explicitly encodes domain knowledge (e.g., GRI indicators), preserves provenance, and combines semantic with symbolic reasoning.

This paper focuses on document-grounded question answering over sustainability reports and financial statements of Vietnamese banks. We emphasize (i) structuring ESG–finance evidence into a consistent KG schema and (ii) KG-guided retrieval to improve multi-hop evidence aggregation and explainability for cross-document queries.

3 ESG Question-Answering System

Figure 1 illustrates the overall architecture and key components of the proposed ESG question-answering system. The system comprises three main modules:

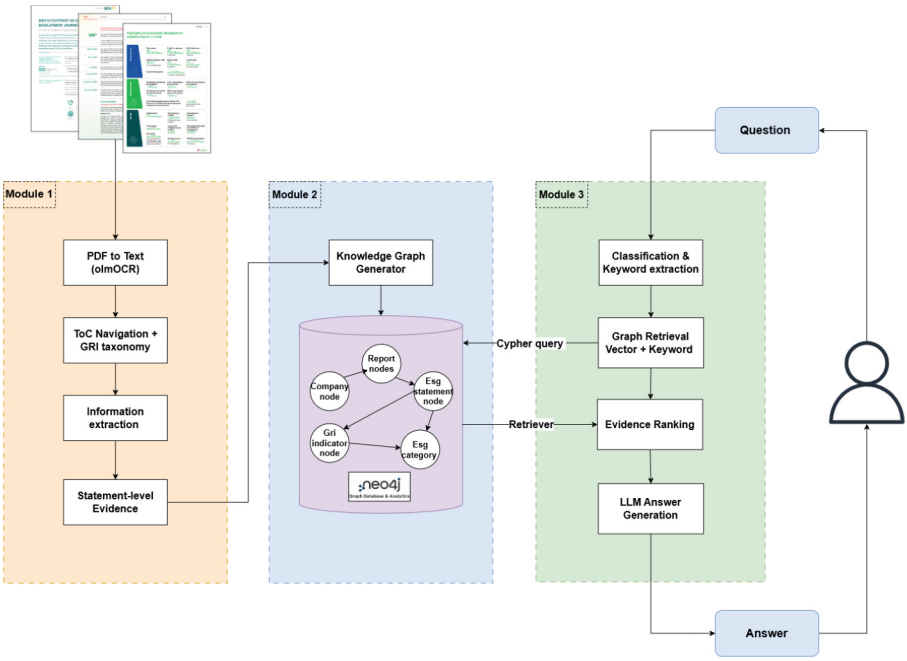


Fig. 1. ESG Question-Answering components.

Module 1 - a report processing module that extracts structured ESG-related information from sustainability and annual reports while incrementally updating the knowledge graph; Module 2 - an ESG knowledge graph construction module that establishes the schema and maintains the curated ESG facts; and Module 3 - a KG-driven retrieval-augmented generation module that retrieves pertinent evidence and generates substantiated responses.

3.1 Report Processing Module

This study presents a report processing pipeline that converts sustainability disclosures from PDF formats into structured representations for knowledge graph integration. We collected sustainability and annual reports from 18 commercial banks in Vietnam, issued during 2023 and 2024. The data collection phase was aimed at creating a representative corpus of ESG practices in the Vietnamese banking sector, emphasizing reports rich in sustainability-related content (e.g., sections on ESG, responsible lending, human capital, and governance). To alleviate noise resulting from formatting differences among various banks, all reports were subjected to document-level normalization, encompassing the standardization of file formats, naming conventions, and the incorporation of fundamental metadata (bank name, publication year, report type).

A key challenge in ESG report processing is that relevant disclosures are scattered across sections and mixed with non-ESG content. To improve extraction

accuracy and efficiency, we use a navigation method that combines the table of contents (ToC) with the GRI taxonomy. The ToC helps identify sustainability-related page ranges and prioritize sections likely to contain standardized disclosures (e.g., Sustainability/ESG, Environment, and GRI Content Index). In parallel, the GRI taxonomy provides a standard backbone for mapping retrieved evidence to GRI indicators, enabling consistent topic standardization and cross-bank comparability. We use the ToC as a soft navigation prior rather than a hard filter: when page titles, headings, or extracted text do not fully align with the ToC, the page is still retained for downstream screening. This design is important for noisy reports because scanned pages, decorative layouts, and multi-column formatting can introduce OCR artifacts or incomplete section boundaries.

We use `olmOCR` to process layout-heavy reports because it preserves reading order and structural cues such as headings, lists, and tables, which are essential for linking extracted statements to section titles and page references [19]. This is particularly useful when the ToC must be reconciled with visually complex pages: the extracted heading sequence provides an additional check that the page content matches the expected section, while table linearization helps retain quantitative disclosures that would otherwise be lost with plain text extraction. We then use the Qwen3 4B instruct model to perform information extraction on the structured OCR output [22]. Rather than a single-pass extraction approach, we implemented instruction prompting to guide the model through schema-constrained extraction tasks [24]. The model is instructed to:

- **Determine Page-level ESG Relevance:** Use the ToC and section information as context to identify pages likely to contain substantive ESG disclosures.
- **Extract Statement-level Evidence:** Identify atomic units of evidence at the sentence or short-paragraph level, including both text and table-derived content.
- **Assign ESG Categories:** Classify evidence into the three pillars (E, S, G).
- **Map to GRI Indicators:** Align evidence with GRI indicators based on disclosure descriptions in the taxonomy. Evidence is either "unmapped" for post-processing or mapped at the topic level when it is too ambiguous for a specific indicator.

Handling Noisy OCR Output: Sustainability reports frequently contain scanned pages, decorative covers, multi-column infographics, and image-based tables, all of which can produce broken reading order, missing characters, or merged cells. Our pipeline mitigates these issues at three stages. (i) *Soft ToC navigation*: pages whose extracted headings only partially match the ToC are retained rather than discarded, so a noisy heading does not cause a relevant page to be skipped. (ii) *Structure-aware extraction*: `olmOCR` preserves headings, lists, and linearized tables, allowing the LLM extractor to recover quantitative disclosures (e.g., GHG figures, training hours) that pure-text OCR would lose. (iii) *Schema-constrained validation*: each candidate `ESG_Statement` must carry

a non-empty `page` reference and a category label; statements that fail validation are flagged as `unmapped` and reviewed in a post-processing pass instead of being silently dropped. In our corpus, this combined strategy recovered evidence on pages where standard text-only PDF parsing returned empty or fragmented output, particularly for layout-heavy infographic pages summarising key sustainability metrics.

3.2 ESG Knowledge Graph Module

We build an ESG knowledge graph (ESG-KG) to transform unstructured ESG content from Vietnamese bank reports into a structured, queryable representation. Knowledge graphs are entity–relation structures that enable data integration, querying, and reasoning across heterogeneous sources [17]. Our ESG knowledge graph connects the information we gather from bank reports to sustainability topics set by GRI [5]. In addition, the graph structure is informed by existing ESG taxonomy enrichment efforts, such as FinSim4-ESG [10]. The resulting graph allows us to track information from documents and combine data at the indicator level for further analysis and improved information retrieval. The ESG-KG schema consists of five main node types:

- `Company`: represents each bank or organization, identified by `company_id` and `company_name`.
- `Report`: represents a sustainability or annual report of a company in a given year, with attributes `report_id`, `company_id`, `type_of_report`, and `year`.
- `ESG_Statement`: represents an ESG Statement extracted from a report, typically a sentence or short paragraph. Each statement includes `statement_id`, `title`, `content`, `embedding_content`, and `page`. The `embedding_content` is used for semantic retrieval in the RAG pipeline [12,14].
- `GRI_Indicator`: represents a specific disclosure topic defined by the GRI framework, with attributes `indicator_id`, `indicator_name`, `GRI_disclosure_detail`, and `topic_title` [5].
- `ESG_Category`: represents the ESG values {E, S, G}.

We instantiate the ESG-KG in Neo4j following the schema illustrated in Fig. 2. This design ensures end-to-end traceability from high-level entities to document-grounded evidence units and aligned sustainability topics.

For each `Report`, we segment the document into candidate evidence units and materialize them as `ESG_Statement` nodes. We then categorize `ESG_Statement` into `ESG_Category` and indicator alignment to `GRI_Indicator`. This LLM-assisted information extraction approach is consistent with recent evidence that iterative prompting and LLM-guided pipelines can improve the construction of schema-conformant knowledge graphs [18,20]. Importantly, all nodes and edges retain provenance (report metadata and page references) to support verification and error analysis. These derived signals support improved retrieval by prioritizing indicators and categories with stronger, multi-evidence support during KG-driven RAG [12].

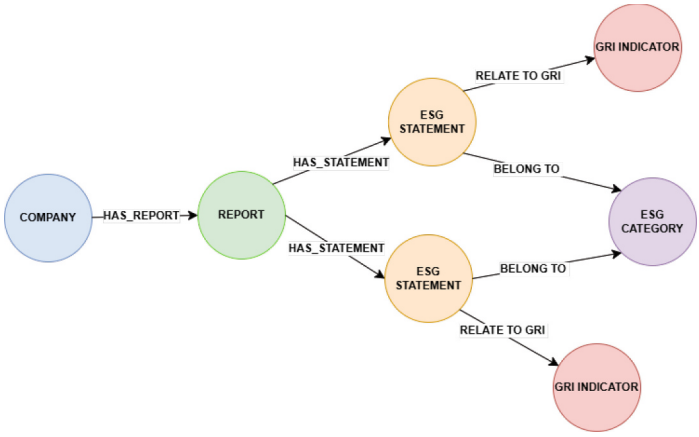


Fig. 2. ESG Knowledge Graph Schema.

3.3 KG-Driven Retrieval-Augmented Generation Module

This module implements a KG-RAG pipeline. Given a user question q, the system employs, (i) a hybrid retrieval strategy (Algorithm 1) that combines graph-based filtering with semantic and keyword-based scoring to retrieve the most relevant evidence from the ESG knowledge graph. The retrieved evidence is then passed to (ii), an answer generator that produces substantiated responses grounded in the provenance-linked statements, ensuring traceability and explainability for end-users and auditors (Fig. 3).

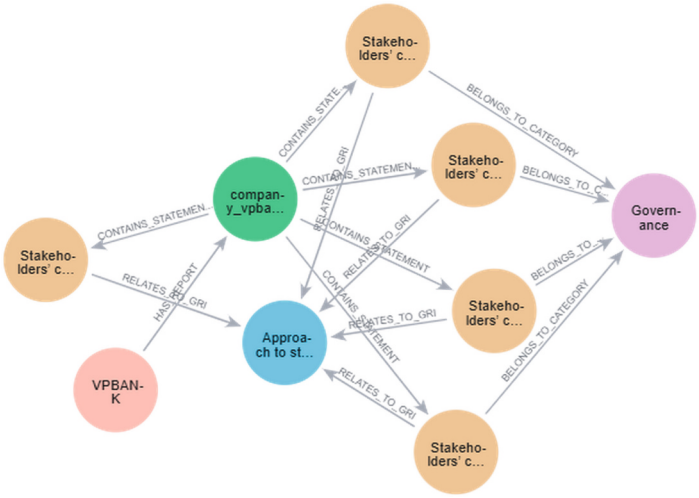


Fig. 3. Illustrating entities and their relationships in the ESG-KG.

KG Retriever: We represent the ESG knowledge graph as $G = (V, E)$, where V is the set of nodes and E is the set of edges. Each node $v \in V$ has a label (e.g., `Company`, `Report`, `ESG_Statement`, `GRI_Indicator`, `ESG_Category`) and a set of attributes that may vary by label. Given a question q, the retriever returns the top-k most relevant `ESG_Statement` nodes as evidence.

Statement Embedding: For semantic retrieval, we encode the `embedding_content` attribute of each `ESG_Statement` into a dense vector using a Qwen-based embedding model [22]. This yields a fixed-size representation $\mathbf{e}_s$ for each statement s, which is used to compute similarity with the query embedding.

Hybrid Graph-Aware Retrieval: We use a hybrid strategy that combines graph filtering with semantic and keyword-based scoring. First, an LLM assigns q to an ESG dimension (E/S/G). This label is used to generate a schema-constrained Cypher query that selects a candidate set of `ESG_Statement` nodes. Next, we score candidates by (i) vector similarity between the query embedding and statement embeddings and (ii) keyword matching for critical entities such as years, metrics, or GRI codes. Finally, we fuse the top results from both scoring streams to form the evidence context.

Answer Generation: The answer generator takes as input the question q and the retrieved evidence S_{context}. We use a fixed instruction prompt to constrain the LLM to produce answers based only on the provided statements. Each retrieved statement is passed together with its provenance (e.g., report name, page number and topic relevant) to support verification and traceability.

Algorithm 1. Hybrid Graph-based ESG Retrieval Strategy

Input: Question q, Knowledge Graph $G = (V, E)$, LLM M_{llm}, Embedding Model M_{emb}
Parameters: Vector Top-k ($k_v = 3$), Keyword Top-k ($k_w = 2$)
Output: Evidence set S_{context}

$S_{\text{context}} \leftarrow \emptyset$
$c \leftarrow \text{Classify}(q)$ using M_{llm} ▷ $c \in \{E, S, G\}$
Step 1: Keyword Extraction
$K_{\text{num}} \leftarrow \text{RegexExtractNumbers}(q)$ ▷ years, percentages, units
$K_{\text{txt}} \leftarrow \text{LLMExtractKeywords}(q, M_{\text{llm}})$ ▷ entities, topics
Step 2: Candidate Generation via Graph
$Q_{\text{cypher}} \leftarrow \text{GenerateCypher}(q, c, G.\text{Schema})$
$S_{\text{cand}} \leftarrow \text{ExecuteCypher}(G, Q_{\text{cypher}})$
if S_{cand} is empty **then**
    $S_{\text{cand}} \leftarrow \text{GetStatementsByCategory}(G, c)$ ▷ fallback
end if
Step 3: Scoring & Ranking
$\mathbf{e}_q \leftarrow M_{\text{emb}}.\text{encode}(q)$
for each statement $s \in S_{\text{cand}}$ **do**
    $s.\text{score}_{\text{vec}} \leftarrow \text{CosineSimilarity}(\mathbf{e}_q, \mathbf{e}_s)$

$s.\text{score}_{\text{kw}} \leftarrow 0$
for $k \in K_{\text{num}}$ **do**
  if $k \in s.\text{content} \vee k \in s.\text{title}$ **then** $s.\text{score}_{\text{kw}} \leftarrow s.\text{score}_{\text{kw}} + 2$
  end if
end for
for $k \in K_{\text{txt}}$ **do**
  if $k \in s.\text{content} \vee k \in s.\text{title}$ **then** $s.\text{score}_{\text{kw}} \leftarrow s.\text{score}_{\text{kw}} + 1$
  end if
end for
end for
Step 4: Hybrid Fusion
$L_{\text{vec}} \leftarrow \text{SortDesc}(S_{\text{cand}}, \text{score}_{\text{vec}})[1 : k_v]$
$L_{\text{kw}} \leftarrow \text{SortDesc}(S_{\text{cand}}, \text{score}_{\text{kw}})[1 : k_w]$
$S_{\text{context}} \leftarrow \text{Unique}(L_{\text{vec}} \cup L_{\text{kw}})$
return S_{context}

4 Experiments and Results

We construct an ESG knowledge graph covering 18 Vietnamese commercial banks. Among them, 10 banks provide sustainability reports for both 2023 and 2024, while 8 banks provide Annual Reports for 2023 and 2024 that include sustainability-related sections. The reports have an average length of approximately 60 pages per document, with the longest Sustainability Report containing 80 pages and the shortest containing 28 pages (e.g., Fig. 4).



Fig. 4. Sample-VPBank Sustainability Report 2024.

To evaluate the proposed KG-RAG framework, we created a question set of 120 ESG questions with support from ESG domain experts. The experts further categorized the questions into three groups: Environmental (55 questions), Social (32 questions), and Governance (33 questions). Since evaluating all 120 questions across all 18 banks would be computationally expensive, we randomly selected 6 banks for evaluation, comprising 10 sustainability reports and 2 annual reports, with report lengths ranging from 28 to 80 pages. Answers generated by different methods were manually assessed by domain experts.

Our experimental design compares multiple retrieval configurations paired with different large language models to assess the contribution of each component to system performance. For each question-document pair, we retrieve the top-3 candidate statements and pass them to the answer generator. Each generated answer is then independently evaluated by domain experts for correctness based on whether it accurately reflects the document evidence and addresses the question. We measure performance by accuracy—the percentage of correctly answered questions—aggregated across ESG dimensions (Environment, Social, Governance) and overall. This setup enables us to isolate the impact of retrieval strategy and LLM choice on the final QA performance.

The expert-led evaluation protocol offers high-quality judgment for a specialized ESG setting, but it is not easily scalable to larger cross-industry benchmarks. In practice, we view this benchmark as a carefully controlled first-stage evaluation. A larger-scale benchmark could combine expert review for a representative subset with lighter-weight validation procedures, such as adjudication guidelines, evidence overlap checks, citation quality scores, and automated faithfulness metrics for answers grounded in retrieved passages. This hybrid protocol would preserve domain rigor while reducing the annotation burden.

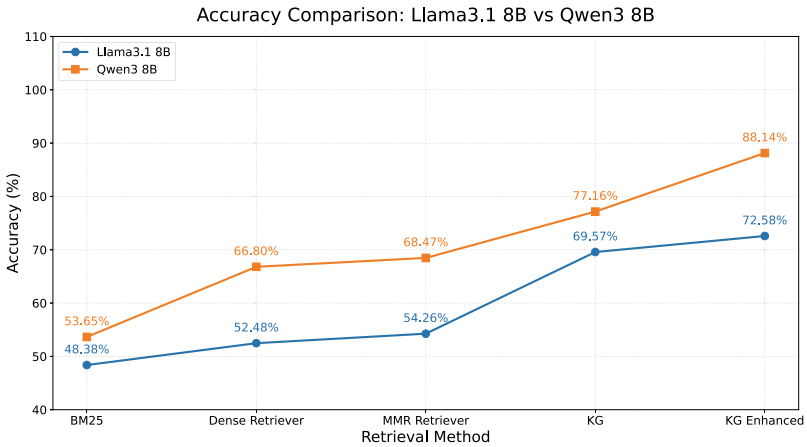


Fig. 5. Retriever wise Accuracy (in percentage) plot for Llama3 & Qwen3.

We experiment with two graph configurations: `KG` and `KG_Enhanced`. The `KG_Enhanced` variant augments the graph-based retriever with a hybrid search component that incorporates keyword matching. As baselines, we compare against three retrieval methods that return the top-3 statements for each question: (i) `BM25`, which ranks evidence using term frequency and inverse document frequency; (ii) `MMR`, which promotes both relevance and diversity to reduce redundancy, using Qwen3-4B embeddings; and (iii) `Dense Retrieval`, which ranks candidates by embedding similarity computed with Qwen3-4B embeddings [22].

For answer generation, we evaluate two LLMs: `Llama3.1-8B` and `Qwen3-8B` [22,25]. Figure 5 reports retriever-wise accuracy (percentage of correctly answered questions) under different retrieval strategies. Overall, both Llama3 and Qwen3 show a consistent improvement when moving from traditional retrieval baselines (e.g., BM25 and MMR) to KG-driven retrieval. In particular, `KG_Enhanced` achieves the best performance across settings, indicating that combining KG filtering with hybrid (semantic + keyword) retrieval improves evidence selection and leads to more accurate, report-grounded answers. These results highlight the benefit of leveraging structured graph constraints for ESG question answering.

We use `Qwen3-8B` as the main answer generator because it provides a practical balance between reasoning quality, context utilization, and computational cost for our document-grounded setting. A larger model may improve difficult cases involving numeric reconciliation or long-range evidence aggregation, but it would also increase inference cost and reduce reproducibility for a pipeline that already includes OCR, retrieval, and KG construction. More importantly, the remaining errors suggest that model scale alone does not solve the core bottlenecks of this task: some failures originate from imperfect evidence extraction, noisy layouts, or incomplete retrieval chains. On this task, the main limitations of `Qwen3-8B` are its tendency to over-commit when evidence is partial, its uneven handling of tabular numeric evidence, and its sensitivity to contexts that contain several near-duplicate statements.

Computational Cost: We measured the runtime of each pipeline stage on a single NVIDIA RTX 3090 (24 GB) GPU. Table 1 summarises the results. Indexing a report (OCR + statement extraction + KG ingestion) takes roughly 4–5 min and is dominated by Qwen3-4B statement extraction over the structured OCR output. After indexing, each query is processed in about 10 s end-to-end, distributed roughly evenly across graph filtering, embedding-based scoring, and answer generation; no single component dominates query-time latency. Because reports are indexed once and then queried many times, this profile is practical for batch auditing workflows and periodic ESG assessment on commodity hardware.

Table 2 reports the accuracy (percentage of correctly answered questions) of Llama3.1-8B and Qwen3-8B under two retriever settings, `KG` and `KG_Enhanced`, across the Environmental, Social, and Governance categories, as well as overall. The best-performing configuration is `KG_Enhanced_Qwen3-8B`, achieving

Table 1. Average runtime per pipeline stage on a single NVIDIA RTX 3090 (24 GB) GPU.

Stage	Avg. time
olmOCR (PDF → structured text)	∼1 min/report
Statement extraction (Qwen3-4B)	∼3 min/report
KG ingestion (Neo4j)	∼30 s/report
Graph filtering (Cypher generation + execution)	∼3 s/query
Query embedding + cosine scoring	∼5 s/query
Answer generation (Qwen3-8B)	∼3 s/query
End-to-end query latency (KG_Enhanced)	∼10 s/query

Table 2. Performance Comparison of Llama3.1 and Qwen3

Category	Environment	Social	Governance	Overall
KG_Llama3.1-8B	68.65%	69.60%	70.45%	69.57%
KG_Qwen3-8B	70.62%	82.23%	78.63%	77.16%
KG_Enhanced_Llama3.1-8B	70.35%	76.82%	70.56%	72.58%
KG_Enhanced_Qwen3-8B	**83.74%**	**92.43%**	**88.25%**	**88.14%**

83.74% (Environment), **92.43%** (Social), and **88.25%** (Governance), which results in the highest overall accuracy of **88.14%**.

Under the `KG` retriever, `KG_Qwen3-8B` outperforms `KG_Llama3.1-8B` across all categories and overall (77.16% vs. 69.57%), with a particularly strong gain on Social questions (82.23% vs. 69.60%). Adding hybrid retrieval (`KG_Enhanced`) improves performance for both LLMs. In particular, Llama3.1-8B increases from 69.57% to 72.58% overall, while Qwen3-8B improves substantially from 77.16% to 88.14%. These results suggest that combining KG-based candidate filtering with hybrid (semantic + keyword) ranking yields more relevant evidence and that Qwen3-8B benefits the most from the enhanced retrieval context. Some examples of retrieved evidence and generated answers are presented in Appendix A to illustrate system behavior.

The values $k_v = 3$ and $k_w = 2$ in Algorithm 1 were selected to balance two complementary signals: k_v should be large enough to cover the descriptive, multi-aspect nature of ESG narratives (where a single question often spans policy, action, and quantitative outcome), while k_w should be small enough to inject high-precision keyword matches (e.g., specific years or GRI codes) without overwhelming the context with redundant numeric mentions. We confirm this design choice empirically with the sensitivity analysis in Table 3.

Table 3 reports a sensitivity analysis of the hybrid retriever for the best-performing configuration, `KG_Enhanced_Qwen3-8B`. The setting $k_v = 3$ and $k_w = 2$ achieves the highest overall accuracy of **88.14%**. Reducing the context to $(2, 1)$ lowers accuracy to 78.92%, indicating that some questions require more

Table 3. Sensitivity of k_v and k_w for `KG_Enhanced_Qwen3-8B`.

(k_v, k_w)	$(2,1)$	$(2,2)$	$(3,2)$	$(4,2)$	$(3,3)$
Overall accuracy	78.92%	82.16%	**88.14%**	86.23%	84.71%

than two supporting statements. Increasing either semantic or keyword evidence beyond the selected setting also reduces performance, with 86.23% for $(4,2)$ and 84.71% for $(3,3)$. This suggests that additional retrieved statements introduce redundancy and make the answer context less focused. Overall, the curve is unimodal around $(3,2)$, supporting it as a robust default for this domain rather than a hyperparameter overfit to a single split.

Table 4. Distribution of error categories on incorrect cases

Error category	Environmental	Social	Governance	Overall
E1: Annotation drift	18%	25%	21%	22%
E2: Distractor-rich chunks	26%	12%	19%	18%
E3: Multi-fact composition	31%	19%	22%	24%
E4: Cross-entity contamination	9%	11%	16%	12%
E5: Implicit reasoning	16%	33%	22%	24%

Incorrect predictions account for 11.9% of all evaluated question–document pairs. We manually reviewed the errors produced by `KG_Enhanced_Qwen3-8B` and grouped them into five dominant categories. Table 4 shows that the largest sources of error are multi-fact composition and implicit reasoning, both at 24% overall. To make these categories concrete:

- **E1 (Annotation drift, 22%):** the model returns a factually correct answer that disagrees with the gold label because of evolving disclosures (e.g., a 2024 figure superseding a 2023 one cited in the question).
- **E2 (Distractor-rich chunks, 18%):** the retrieved page mixes the target metric with several similar values (e.g., training hours by gender, by level, by region), and the LLM picks the wrong one.
- **E3 (Multi-fact composition, 24%):** the answer requires aggregating values from two or more pages (e.g., total Scope 1+2+3 emissions reported in separate tables); strongest in Environmental questions (31%).
- **E4 (Cross-entity contamination, 12%):** statements about a parent group or subsidiary are confused with the target bank.
- **E5 (Implicit reasoning, 24%):** the evidence implies the answer (e.g., a stated policy implies a yes/no on a related practice) but does not state it literally; strongest in Social questions (33%).

These patterns suggest concrete improvements: E2 and E4 are addressable by tighter graph filtering on `Company` and `topic_title` edges; E3 calls for explicit multi-hop traversal over `HAS_STATEMENT` chains rather than a single retrieval pass; and E5 would benefit from a reasoning-oriented prompt or a chain-of-thought generator. Annotation drift (E1) is partly an artefact of the benchmark and motivates the hybrid evaluation protocol discussed below. Cross-entity contamination (12%) being the lowest category indicates that KG filtering already removes most off-scope matches.

5 Limitations and Generalizability

Model-Level Limitations: `Qwen3-8B` is a pragmatic rather than upper-bound choice: it offers good answer quality at manageable cost but remains limited in long-range evidence synthesis, numeric reconciliation across tables, and calibrated abstention when retrieval is incomplete. The error patterns in Table 4 reflect these limitations directly, especially for E3 (multi-fact composition) and E5 (implicit reasoning).

Data-Level Limitations: The current benchmark covers 18 Vietnamese commercial banks and 120 expert-curated questions, narrower than the broader space of sustainability reporting practice. Extraction quality also remains contingent on OCR fidelity: scanning artefacts, decorative layouts, and image-only tables can still produce missing or fragmented evidence despite the mitigations in Sect. 3.1.

Evaluation-Level Limitations: Manual expert assessment provides high-quality judgment but is hard to scale. We view the current setup as a controlled first-stage benchmark; a hybrid protocol that combines expert adjudication on a representative subset with automated metrics (faithfulness, evidence overlap, citation precision/recall) would scale evaluation to larger cross-industry corpora

6 Conclusion

This paper presented a knowledge-graph-augmented RAG system for ESG question answering over Vietnamese banking reports. The pipeline converts layout-heavy PDFs into structured evidence using `olmOCR`, extracts statement-level disclosures with LLM-guided prompting, and organises them into an ESG knowledge graph aligned with the GRI taxonomy. Building on this graph, we introduced a KG-driven retriever with hybrid (semantic + keyword) scoring that returns provenance-preserving evidence for answer generation. Experiments on expert-designed ESG questions show that KG-based retrieval—particularly the enhanced hybrid variant—substantially improves accuracy over BM25, MMR, and dense baselines (88.14% vs. 53–69%), confirming the value of structured constraints and evidence traceability for ESG analytics.

As discussed in Sect. 5, the framework is modular and we view it as a transferable blueprint for traceable, document-grounded QA in domains where narrative claims, numeric evidence, and reporting standards must be connected explicitly. Future work will extend the benchmark to additional sectors and integrate automated faithfulness metrics into a hybrid expert–automatic evaluation protocol.

A Appendix

Question: Did female employees hold more than 50% of managerial positions at ACB in 2023? **(S)**
Answer: Yes. Female employees held more than 50% of managerial positions at ACB in 2023. Specifically, 51% of managerial-level employees were female (an increase of 2% compared to 2022).
Retrieved Reports:
- ACB Sustainability Report 2023 – Page 39 (GRI 405-1: Diversity and Equal Opportunity)

Question: Did ACB reduce its total GHG emissions across all three scopes in 2023 compared to 2022? **(E)**
Answer: Yes. Total GHG emissions decreased by 9%, from 34,355 tCO_2e (2022) to 31,238 tCO_2e (2023).
Retrieved Reports:
- ACB Sustainability Report 2023 – Page 33 (GHG emissions summary; reporting across scopes)

Question: Does ACB have a sustainable credit package with a limit of at least VND 2,000 billion in 2023? **(G)**
Answer: Yes. ACB introduced the *Green and Social Credit* package in 2023 with a limit of at least VND 2,000 billion, described in the Chairperson's Message.
Retrieved Reports:
- ACB Sustainability Report 2023 – Page 8 (Chairperson's Message; Green and Social Credit package)

Question: What was the average number of training hours per employee at ACB in 2023? **(S)**
Answer: The average number of training hours per employee in 2023 was **66 hours**, based on a total of 856,831 training hours reported for the workforce.
Retrieved Reports:
- ACB Sustainability Report 2023 – Page 36 (GRI 404-1/404-2: Training and Education)

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